

1 Incentive Wage Calculation - Process

Basics

The HYDRA Incentive Wage Calculation uses the data that has been recorded by a BDE or PZW system. For the different systems of premium or incentive wage, the application uses different subsets of the shop floor data recorded to calculate the incentive wages.

- Postings of operations and persons recorded by the Shop Floor Data Collection BDE in combination with the posting of times to the different resource performance accounts (RPA) performed by the Machine Data Collection MDE.
- Approved bonuses and deductions recorded by the Premium and Incentive Wage LLE
- Changes of group recorded by the Premium and Incentive Wage LLE
- Attendance, absence and break times of the employees recorded by the Personnel Time Management PZW

In addition to the transaction data above, the master data stored in the system for orders and persons and the LLE payment rules affect the data processing for the calculation of wages. For example, a configuration in the master data releases specific persons and orders for the processing in the incentive wage and bonus calculation.

Another important component of the incentive wage calculation is the complete (time) data collection in the relevant HYDRA program module. The objective is to track the complete time between clock-in and clock-out of an employee and to post the times to the relevant operations in the HYDRA system. To fulfill this requirement, HYDRA provides the waiting period processing where all times that an employee is not logged on to an operation are automatically posted to a waiting period operation.

Multiple systems exist to calculate incentive wages. On one side there are systems to calculate person-related piecework, on the other there are purely group-related bonus systems. The premium system used specifies which data recorded in HYDRA is used to calculate the incentive wage.

Using data of the Shop Floor Data Collection (BDE)

The BDE personnel postings are often used to calculate individual piecework or person-related bonuses.

To calculate group bonuses, the BDE order postings and sometimes the BDE personnel postings specify the key figures used. To distribute group results to the separate persons, the system generally uses the BDE personnel postings.

In the BDE, the following data is automatically transferred from the operation to the BDE order and personnel posting when an OP or a person is logged on (condition: the HYDRA products to calculate premium and incentive wage and/or to calculate group bonuses are licensed):

- **Single piece specification t_e**
Specified in format hours with 3 decimal places per 1000 pieces
- **Setup specification t_r**
Specified in format hours with 3 decimal places
- **Single piece specification for the production resource t_{eb}**
Specified in format hours with 3 decimal places per 1000 pieces
- **Setup specification for the production resource t_{rb}**
Specified in format hours with 3 decimal places
- **Wage type**
The wage type often specifies how a data record is used for the incentive wage calculation. Different wage types exist for piecework, bonus wage, time wage, overhead costs or other.
- **Premium group**
If the HYDRA product to calculate group bonuses is licensed, then the premium group assigned to the machine is transferred to the order and personnel postings. Optionally and as part of a customization, the premium group can be entered on the BDE terminal when the order is logged on.

In rare cases of the incentive wage calculation, the operator position (function) of the person at the machine is relevant. If required, this operator position can be entered and then processed when the person is logged on. This is a custom function. Another special feature of the incentive wage calculation can be that the person can enter a premium indicator for the activity performed when the person logs on. This premium indicator is then used to calculate the formula-based premium wage. This functionality is implemented via customization.

Using data of the Personnel Time Management (PZW)

In seldom cases, the data of the Personnel Time Management (PZW) is used to calculate the incentive wage. You can use PZW wage type postings or the totaled work day result to calculate the piecework wage or an individual bonus.

In some cases, the PZW wage type postings are assigned to the premium groups via "Change of group". The PZW wage type postings are then used to calculate the key figures for the group bonus and/or to distribute the group results to the persons.

Using data of the Premium and Incentive Wage (LLE)

On the office client, you can optionally record bonuses and changes of group (additional license or customization) for the calculation of the Premium and Incentive Wage (LLE). It depends on the system used to calculate the incentive wage whether bonuses or changes of group are included in the calculation.

You can use bonuses to manually change the premium calculation.

You can use the changes of group to assign the person-related data records of the PZW or the BDE to premium groups.

Data processing

The postings of the employees for the orders they have processed are logged in the HYDRA system. On a daily basis, these postings are transferred to the data of the incentive wage calculation to calculate the wages. Because of the above mechanism, the incentive wage calculation only includes data up to the previous day because the attendance times of the current day are only available on the next day.

If you want to correct wage-relevant data, you directly use the editing functions of the shop floor data collection and the personnel time management and you therefore directly change the output data. The incentive wage calculation then processes the data with the next run of the wage calculation (usually on the next day). The data recorded in the HYDRA system, e.g. BDE personnel postings, is only stored in the system for a specified time (35 days by default). You can only make changes to this data during this time.

For the HYDRA Premium and Incentive Wage, you can configure a longer period for the retention of wage-relevant data on the HYDRA server. If you need a longer period of data retention, these settings can be changed in the system. If you are not sure about the configuration of the data management, contact MPDV.

Overview

The wage calculation is the core function of the Premium and Incentive Wage. The system uses the BDE and PZW postings of the employees to calculate the results for the time tickets or the work day results of the persons and premium groups.

There are two options to start the wage calculation:

1. Use the HYDRA scheduler to start a daily wage calculation that calculates all work day results for persons and premium groups up to the previous day.
2. Manually start the calculation via GUI for specified days and persons or premium groups. For more information, refer to the documentation of this application on the client.

You can repeat the wage calculations as often as you like. Make sure that the data of the relevant day is still available in the system.

For further information, refer to the following documents:

- Information on the [settlement day](#).
- Information on the [performance efficiency calculation](#).
- Information on the [group bonus and premium areas](#).

Automatic recalculation of historical data

In mode *Required days until*, the wage calculation automatically recalculates all day results for persons and premium groups up to the date transferred. To this end, the system memorizes whether changes have been made to data of persons or premium groups. The relevant day results for persons and premium groups must then be recalculated. Also the automatic wage calculation entered in the scheduler starts in this mode and recalculates all relevant data up to the day before today.

The logs of the automatic wage calculations entered in the scheduler are recorded in the HYDRA system logs as application HYL_COMPUT. The manual wage calculation also creates entries here. The column *Number of data records* shows the sum total of the calculated day results for persons and premium groups. It provides an approximate evaluation of the volume that must be calculated.

The following activities trigger an automatic recalculation for the relevant days:

- Editing of bonuses (also approval/rejection) for the relevant person or premium group on the specified day
- Editing of PZW wage type postings (also approval/rejection) for the relevant person on the specified day (also if the data has been passed from external systems via interfaces)
- Recalculation of day results by the PZW labor time calculation for the relevant persons on the specified day
- Editing of BDE postings using the editing function for order-related postings of the relevant person or premium group on the specified day

Some editing activities do **not** trigger an automatic recalculation because very long periods of time or very large data volumes might be affected. In these cases, the user must manually start a wage calculation in mode *Period* for the relevant persons or premium groups, if required. Examples of such activities:

- Editing of changes of groups (these can cover a very long period)
- Editing of master data of premium groups (a period cannot be limited effectively).
- Editing of assignments of premium groups to premium areas (a period cannot be limited effectively).

Combining consecutive time tickets

This function is used to combine BDE personnel postings that have been interrupted by an automatic shift change. This is required in case of an overtime after end of shift. With a manual recording of quantities, the problem is then that the first part of the posting has a duration, but no quantity. The second part of the posting then includes the complete quantity recorded with staff logoff, but includes only a short duration. If the wage calculation calculates each of these postings separately, then the resulting performance efficiency rates would be very low with posting 1 and very high with posting 2.

To avoid this problem, the wage calculation can combine similar BDE personnel postings that are directly one after the other in time and consider them as one posting and then create one time ticket.

To activate the function to combine consecutive time tickets, go to the *Basic settings for incentive wage*.

The following conditions must be fulfilled to combine two BDE personnel postings:

Column	Condition
Machine	must be the same
Person	must be the same
Order/operation	must be the same
Cost center	must be the same
Quantity units	must be the same
Times of postings	Logoff time of the first posting must be identical to the logon time of the second posting.
Wage specifications t_e , t_r , t_{eb} , t_{rb}	must be the same
Wage type	must be the same
Premium group •	must be the same
Operator position/function	must be the same
Premium indicator	must be the same
Performance units	must be the same
User fields	Numeric user fields must be 0 or empty with both postings. Alphanumeric user fields must be identical in the first and second data record.

When two BDE postings are combined, the data is either totaled or the data of the second posting is used:

Column	Action
Quantities	are totaled.
Quantity reasons	of the second posting are used. (The second posting includes the

	information on the event that triggers the logoff).
Times of postings	are identified using the logon time of the first posting and the logoff time of the second posting.
Shift information	of the second posting is used.
Durations and resource performance accounts	are totaled.
Interruption reason	of the second posting is used. (The second posting includes the information on the event that triggers the logoff).
Performances	are totaled.